

### **BIOSTATISTICS MANDATE: CUSTOM WEIGHTED DICE (CWD)**

For this clinical trial, the standard Intersection over Union (IoU) is invalid. You must calculate the Custom Weighted Dice (CWD) coefficient for each patient's segmentation mask.

Parse the decompressed binary file to extract the Ground Truth Pixels (GT) and Predicted Pixels (Pred).

Determine the absolute minimum value between GT and Pred. This represents the intersection.

Extract the specific etiology weight from the hidden sheet of the Excel file.

Apply this exact formula:

$$CWD = \text{Weight} \times \frac{2 \times \min(\text{GT}, \text{Pred})}{\text{GT} + \text{Pred}}$$

*(Calculate all CWD values to exactly two decimal places).*